



Operating Systems: Introduction

Explore the basics of operating systems and the core features that power your computer.

45 min 115,904

Start AttackBox

Save Room

591 Recommend

Connectivity

Options

Room progress: 11%

Task 1 Introduction

You turn on your phone or laptop every day, and everything just works: apps launch, files open, music plays, and the whole system feels seamless. But what actually makes all this possible? In the *Computer Fundamentals* module, you explored the physical components of a computer and the different types of computing devices. Now it's time to uncover the invisible layer that ties it all together: the **operating system**, or **OS**.



Scenario

Your friend just upgraded their primary computer and, knowing you're getting into tech, has generously gifted you their old machine. It's been sitting untouched for a while, and he can't remember much about it, only that it "used to run well". Before you can decide whether to upgrade it, wipe it, sell it, or turn it into your next project, you need to figure out precisely what you're dealing with.

Learning Objectives

- Understand what an operating system is and the role it plays
- Explain the core duties of an operating system
- Identify common OS types and their typical use cases
- Practice interacting with an OS to gather system information

Prerequisites

Before you begin this room, it is recommended that you complete the *Computer Fundamentals* module (coming soon) of the *Pre Security* Path. This room builds on core ideas introduced earlier, especially your understanding of computer hardware, device types, and how modern systems boot and operate.

- Inside a Computer System
- Computer Types

Answer the questions below

I understand the learning objectives and am ready to learn about operating systems!

No answer needed

✓ Correct Answer

An **operating system** (OS) is the core software that coordinates everything happening on a computer. It sits between the user, applications, and the system's physical hardware, acting as the invisible manager that keeps the entire machine running as one unified system.



A helpful analogy is to think of your computer as a busy airport, with all its components functioning together.

- Your **hardware** (CPU, RAM, storage, connected devices): The runways, airplanes, fuel systems, radar, and other physical infrastructure.
- Your **applications** (web browser, game launcher): The various airlines and their passengers, all trying to take off, land, and request services.
- Your **operating system** (Windows, Linux, macOS): The entire air traffic control system, directing all of this activity. It schedules resources, manages traffic, resolves conflicts, and ensures safety.



We need an operating system because it provides this all-important job of coordination and structuring that makes modern computing possible. Without an OS, each application would need direct control over the CPU, memory, files, devices, and security. This would quickly cause conflicts, and the OS handles this by acting as the central organizer. In the Computer Fundamentals module (coming soon), you learned about various computer components and their duties. Now, we will see how the operating system manages and allocates these resources when you use your PC.

System Privilege Layers

Inside a modern computer, different parts of the system operate at various permission levels. Some components can communicate directly with the hardware, while regular applications run in a safer, restricted environment. This separation is intentional and helps prevent conflicts and security issues.

- **Kernel space:** The privileged, locked-down core of the OS. This is where the kernel, the part of the operating system that directly manages hardware and system resources, runs. It has unrestricted access to the CPU, memory, storage, and all hardware components.
- **User space:** Where all standard applications run. Applications in the user space are deliberately prevented from accessing hardware directly. Whenever they need to open or save a file, play a sound, or connect to Wi-Fi, they must make a system call and request that the kernel act on their behalf.

Building on our airport analogy, let's zoom in on the concept of privilege separation. The kernel space is the control tower, a strictly secured area where only trusted air-traffic controllers (the kernel) work. They alone can directly control the runways, radar, and other hardware. Applications in the user space are like airlines and passengers on the ground. They can't enter the tower or touch the equipment. Instead, they radio requests (system calls) to the tower, which handles them safely. This separation keeps the OS reliable: one faulty app can't crash the whole system, just as no airline can operate safely without the tower's control.

Operating System Duties

Now that you know what an operating system is and how system privilege is separated, let's look at what it actually does behind the scenes. Every OS is responsible for a few core duties that allow your computer to run safely, efficiently, and predictably.

OS Responsibility	What the OS Does	Example
Process Management	Creates, schedules, prioritizes, and terminates running programs. The OS decides how much CPU time each process gets, making multitasking feel seamless	Opening multiple apps, like your browser, music player, and social media, without your computer freezing
Memory Management	Allocates RAM to processes, protects the app's memory from other processes, and reclaims memory when apps are closed. When RAM runs low, the OS uses virtual memory to keep your system stable	Opening multiple app at once, the OS allocates RAM to each one and keeps them isolated so they don't interfere or crash each other

File System Management	Organizes files into directories, handles naming, paths, permissions, metadata (name, size, type, timestamps)	Creating a new folder, saving a photo, or setting a file to "read only"
User Management	Handles multiple user accounts, authentication, and permissions to determine who can access what	Logging in with your password and keeping your files inaccessible to other user accounts
Device Management	Loads drivers and provides a universal interface (hardware abstraction layer), so apps can say "print this" or "play this sound"	Plugging in a new mouse, printer, or external hard drive and having it work immediately

Operating System Security

It is important to understand that every OS also acts as a security foundation. Before any antivirus, [firewall](#), or security tool is introduced, the OS is already enforcing protections in the background, some of which we covered above.

At a basic level, your operating system handles

- **Authentication:** Verifies who you are through login passwords and biometrics
- **Permissions:** Controls exactly what each user and app is allowed to read, write, or execute
- **Isolation:** Keeps every process in its own protected box (kernel/user space separation)
- **System Protection:** Safeguards critical system files and settings from unauthorized changes

Getting Hands-on

Now that you have a solid understanding of what an operating system is and its main duties, let's get hands-on with your new computer. Click the **Start Lab Machine** button below to fire up your new computer. The machine will open in split view, at which time you'll gain access to the computer's Desktop. If split view is not opening, press the **Show Split View** button at the top of the page.

Your virtual environment has been set up 🟢

All machine details can be found at the top of the page.



Lab machine ⓘ

Status: ● On

[See connection methods](#)

Remember, your friend said he doesn't know much about the PC he gave you, so let's dive in and try to gain some initial information about what you're working with. On the Desktop of the machine, there is a shortcut named **About This Computer** that opens your computer's **System Monitor**. Go ahead and open it now, as you will need it for some of the questions in this task.

Applications Places System 🔍

System Monitor

Monitor Edit View Help

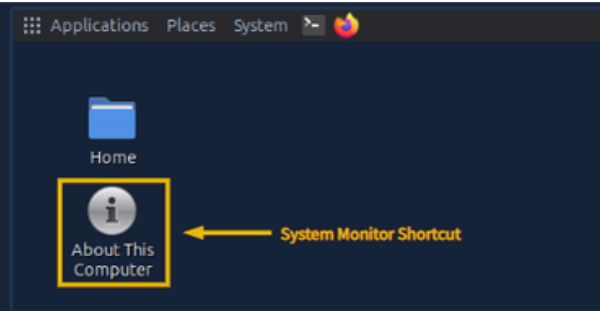
System Processes Resources File Systems



tryhackme
Ubuntu
 Release 24.04.1 LTS (Noble Numbat) 64-bit
 Kernel Linux 6.8.0-1016-aws x86_64
 MATE 1.26.2

Hardware
 Memory: 1.9 GiB
 Processor: AMD EPYC 7571 × 2
 Graphics: llvmpipe (LLVM 17.0.6, 256 bits)

System Status
 Available disk space: 9.1 GB



Answer the questions below

Which OS space has unrestricted access to your computer's hardware?

Kernel Space

✓ Correct Answer

Which OS responsibility manages user accounts, authentication, and permissions?

User Management

✓ Correct Answer

After opening the **About This Computer** shortcut, you are greeted with an overview of the system's specifications.

What version of Ubuntu Mate is your computer running?

1.26.2

✓ Correct Answer

Check out the **Hardware** section of the **System** tab.

How much memory is allocated to your machine?

1.9 GiB

✓ Correct Answer

Task 3 ✓ OS Interaction and Landscape

OS Interfaces

Now that you have a solid understanding of the operating system and its various responsibilities, let's look at how we interact with the OS. Interaction with the OS can be divided into two main parts: the **graphical user interface (GUI)** and the **command-line interface (CLI)**.

Graphical User Interface

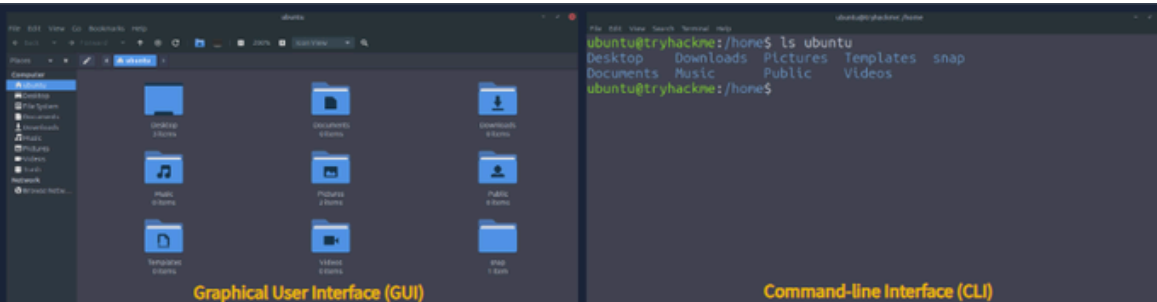
The GUI is what you're most likely used to interacting with. It provides a graphical representation of all the information you want to access on your computer. Think of folder icons, windows for your applications, and menus for settings. We can imagine the analogy of using a navigation application. You tap an icon of the place you want to visit, and the app generates directions for you, eliminating the need for typing.

Command-line Interface

The CLI is where you enter specific text-based commands to retrieve or manipulate information. Instead of clicking on icons, you tell the computer exactly what you want using words and syntax that the system understands. This gives you far more precision, control, and speed, especially for advanced tasks, but it requires familiarity with the commands. Back to the maps analogy. Using the CLI is like entering the exact GPS coordinates of your destination. It's direct and extremely accurate, but only if you know the correct information to type.

Later in this module, you will explore the CLI on both **Linux** and Windows to learn how to navigate files, inspect system information, and interact with your OS beyond the GUI.

In the screenshot below, you can see that the GUI and CLI are both used to retrieve the same information. In this case, to display the contents of the **ubuntu** user's **home** directory. The GUI requires a few clicks for folder navigation, whereas the CLI requires a command to list the directory contents.



The Operating System Landscape

Nice! We are gaining a much clearer picture of the OS, its responsibilities, and how we can interact with it to manage our computer. Now it's time to look at the bigger picture; not all operating systems are the same. Different devices and jobs demand different designs, ranging from your phone to a web server in a data center. Below are the five major categories you'll run into in the real world:

Operating System Type	Primary Use Case	Key Characteristics
Desktop	Personal computers, daily work, gaming, content creation	Rich graphic interface, runs many apps at once, user-focused
Server	Web hosting, databases, cloud services, back-end	Headless (no GUI), maximum uptime, multi-user, remote access
Mobile	Smartphones and tablets	Touch-based UI, power efficient, always connected, app sandboxing
Embedded	Appliances, cars, IoT devices, smart TVs, routers	Tiny footprint, runs on limited hardware
Virtual/Cloud	Lab machines, containers, cloud instances	Lightweight, scalable, rapid deployment

Real World Operating Systems

Now that you've seen the different types of operating systems and what they're designed for, let's look at the major families of operating systems you'll encounter in the real world. Each family fills one or more of the OS types we just explored. To keep things organized, we'll highlight the common versions or distributions you'll see in each and follow the same categories as above: Desktop, Server, Mobile, Embedded, and Virtual/Cloud.

Desktop

- **Windows:** The most widely used operating system on personal computers
Windows 10 (end-of-life), Windows 11
- **macOS:** Apple's desktop OS, known for its polished GUI and integration with other Apple devices
Sonoma (14), Sequoia (15), Tahoe (26)
- **Linux:** Not a single OS but a family of open-source operating systems called distributions
Ubuntu, Debian, Fedora

Server

- **Windows:** Used in large networks, data centers, and corporate environments
Server 2016, 2019, 2022, 2025
- **Linux:** The vast majority of web servers, trusted for its reliability and open-source nature
Ubuntu Server, Debian, CentOS, Red Hat
- **Unix:** Large enterprises, finance, telecom, government
IBM AIX, Oracle Solaris

Mobile

- **Android:** The most widely used mobile OS, which runs on phones, tablets, and smart devices
Android 14 - 16, Manufacturer versions
- **iOS:** Apple's mobile OS running on iPhones, iPads, and other devices
iOS 17, 18, 26

Embedded and IoT Devices

- **Embedded Linux:** Specialized OS built into devices with dedicated functions
OpenWrt, Ubuntu Core, Yocto Project
- **Real-Time OS:** Designed for apps where tasks need guaranteed response times (aircraft controls)
FreeRTOS, VxWorks, QNX

Virtual and Cloud

- **Cloud/VM:** Massive data centers that host websites, apps, and streaming services
Ubuntu LTS, Amazon Linux, Rocky Linux
- **Container-optimized:** Lightweight alternatives to VMs that package just the app and its dependencies
Alpine Linux, Bottlerocket AWS, Flatcar Linux



Why So Many Operating Systems?

Different devices and environments require different capabilities from an OS. A laptop must be user-friendly and support multitasking. Servers require stability, security, and must be able to run continuously without interruption. Mobile devices need power efficiency and hardware integration to extend battery life. Embedded systems use lightweight operating systems designed for a specialized purpose.

The companies and communities that develop these operating systems also have their own goals. Some focus on ease of use, performance, security, openness, or customization. Because each environment values different capabilities, no single OS is the perfect fit for every situation. Instead, an ecosystem of operating systems has evolved.

Continuing Your Investigation

From the previous task, you learned that your new computer is running the Ubuntu distribution of Linux. You were also able to determine the version and release. Let's continue the investigation and gather further information about the system. You will continue to use the **About This Computer** shortcut, then jump into the **Home** directory, which can be found on your computer's Desktop.



ApplicationsPlacesSystem

Fri Aug 28, 16:07

Documents

note.txt (/home/alex/Documents) - Pluma

File Edit View Search Tools Documents Help

Open Save Undo

note.txt x

1 I hope you enjoy the new computer. -Alex
2
3 ...and this flag THM{new_pc_for_free!}

Plain Text Tab Width: 4 Ln 3, Col 39 INS

"note.txt" selected (81 bytes), Free space: 9.1 GB

Answer the questions below

Open the **File Systems** tab in **System Monitor**.
What **Type** is listed for the **/dev/root** device?

ext4

✓ Correct Answer ?

After opening the **Home** directory on the Desktop, how many user directories exist?

3

✓ Correct Answer

Navigate to Alex's home directory and explore the **Documents** folder.
What is the flag value contained in **note.txt**?

THM{new_pc_for_free!}

✓ Correct Answer

Well done, you've reached the end of **Operating Systems Introduction**. In this room, you learned what an OS really does behind the scenes and explored key concepts, including privileges, user management, memory handling, and processes. In the practical exercise, you investigated a mystery computer gifted to you by a friend. You used the operating system to take a peek into the hardware and file systems it helps manage.

Key Terminology

Let's recap the core terms you've learned. These definitions will help solidify your understanding before moving on to further learning.

- **Operating system (OS)**
The core software that manages hardware, applications, and all system resources.
- **Kernel space**
The OS's highly privileged area with direct hardware access, and the home of the kernel, which directly manages hardware and system resources.
- **User space**
The area where regular applications run with limited permissions for safety and system stability.
- **Graphical user interface (GUI)**
The visual part of the OS, windows, icons, and menus, that lets you interact through clicking and tapping.
- **Command-line interface (CLI)**
A text-based interface where you type commands to control the system with precision and speed.

Further Learning

In the following rooms of the Operating System module (coming soon), you'll dive deeper into Windows and Linux operating systems, learning how to navigate each system using both the GUI and CLI. You'll gain further hands-on experience performing real tasks, gathering system information, and interacting with the OS!

- [Windows Basics](#)
- [Linux CLI Basics](#)
- [Windows CLI Basics](#)

Answer the questions below

Complete the room and continue on your cyber learning journey!

No answer needed

✓ Correct Answer



Great work Abhishek26! Room completed!

Your skills are skyrocketing!

Completed tasks

 4

Points earned

 56

Streak

 8

Windows Basics

Learn to navigate Windows, manage files, and use essential system tools.

45 min 100,122

[Save Room](#)[553 Recommend](#)[Connectivity](#)[Options](#) ▼

Room progress: 10%

Active machines information

IP Address: 10.49.166.59 ▼

Task 1 Introduction

Every operating system has its own personality, and if you've used a computer at school, work, or home, there's a good chance you've already met the most familiar one: **Microsoft Windows**. In the previous room, you explored what an operating system is, the behind-the-scenes manager that keeps your device running smoothly. Now it's time to step into a real example and see the previously learned concepts take shape on a system you may already recognize. In this room, you'll get hands-on experience with the Windows interface and begin building practical skills that will lay the foundation for the subsequent rooms of the module.

Scenario

In this room, you'll take on the role of a new employee starting your first day at **TryHatMe**. Once you log in to your workstation, you'll get to know the Windows desktop, learn how the interface works, open tools from the Start menu, and find your way through company folders. Your supervisor will guide you in creating and organizing files, changing system settings, checking the Task Manager, and reviewing basic security settings. By the end of these onboarding tasks, you should feel comfortable using Windows and handling the daily tasks you'll be expected to handle.



Learning Objectives

- Navigate the Windows graphical interface, including the desktop, taskbar, and Start menu
- Use File Explorer to browse folders, understand file paths, and organize files effectively
- Check system settings and personalize the Windows environment using the Settings app
- Use basic system tools like Task Manager and Windows Security to monitor performance and verify system protection

Prerequisites

Before you begin, it is recommended that you complete the following rooms to make sure you have the foundational knowledge needed.

- Work through [Inside a Computer System](#) to learn how computers start up and how components like the [CPU](#), [RAM](#), and storage interact
- Go over [Computer Types](#) to understand the variety of computing systems and how their roles, capabilities, and designs differ
- Complete [Operating Systems: Introduction](#) to build a clear understanding of what an operating system does and how it functions

Machine Access

Click the **Start Lab Machine** button below to launch your Windows Server 2019 workstation. The machine will open in split-screen mode, giving you access to the Windows desktop used throughout this room.

Your virtual environment has been set up ✓

All machine details can be found at the top of the page.



Lab machine ⓘ

Status: ● On

🔗 See connection methods

Answer the questions below

I understand the learning objectives and am ready to learn about Windows!

No answer needed

✓ Correct Answer

Task 2 ✓ Exploring the Windows Workspace

Before Windows had its current polished look, Microsoft's operating systems were much simpler. Early computers ran [MS-DOS](#), which showed a black screen where you had to type commands instead of clicking icons. In 1985, Microsoft released Windows 1.0, a basic graphical user interface ([GUI](#)) built on top of DOS, introducing windows, menus, and mouse controls to personal computers. Over time, Windows has added more features, which turned the original shell into a complete operating system. Today, the modern Windows [OS](#) is the result of those changes. Let's dive into Windows and see how the operating system concepts you learned about in the previous room present themselves in the world's most widely used desktop operating system.

Logging in and Authentication

Before gaining access to the Windows Desktop, you must authenticate (prove your identity) to the system. The authentication process verifies your identity and determines the actions you're allowed to take once logged in. When a Windows system starts, it displays a login screen where a user account must be selected and authenticated with a password, PIN, or another verification method. Each user is assigned a set of permission levels that determine their access to files, settings, and system functionality. Windows commonly uses the account types below.

- **Guest:** A restricted account intended for temporary access, with minimal permissions and no ability to change system settings
- **Standard:** A user account for everyday tasks, such as running applications and changing personal settings, without access to system-wide changes
- **Administrator:** A privileged account with full control over the system, including software installation, configuration changes, and user management

Upon starting up the machine in the previous task, you are automatically logged in using an Administrator account. This will allow you to explore system settings, install programs, and perform administrative actions without being restricted by permission limitations.

The Windows Desktop

In the last room of this module, we explored the comparison of an OS to an airport. Let's continue with this analogy. If you, the user, are a passenger, and the login screen represents the airport security checkpoint, you can consider the Desktop to be the airport terminal. After entering, it is the first area you gain access to, and all other subsequent areas branch out from this point.



Let's have a look at the Windows Desktop and cover some of its core features together. When you first log in, you're presented with two main areas.

- **Desktop:** The main workspace where files, folders, and shortcuts live
- **Taskbar:** A control strip that provides access to applications, system tools, settings, and notifications

Let's break these down further to get a better picture. Please note that the instance used in this room and the screenshot below are using Windows Server 2019. While newer Windows versions may differ slightly, these core components and concepts remain consistent.

1. **Desktop icons:** Shortcuts to items like the Recycle Bin, folders, and frequently used applications. It is fully customizable
2. **Start menu:** Primary way to access applications, settings, and power options. From here, you can log out, restart, or power off your machine
3. **Search:** Quickly find applications, files, folders, and system settings by using keywords
4. **Task View:** Allows you to see all currently open windows and quickly switch between them
5. **Pinned Applications and Folders:** Your most used applications and folders can be pinned here
6. **Network and Audio settings:** This section can be customized to suit your needs
7. **Date and Time:** Opens up to a full calendar. Date and time settings can be accessed here, too
8. **Notifications:** Displays computer or application notifications. Network and other settings can also be accessed

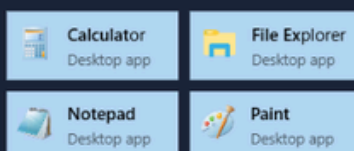


Start Menu

When using Windows, the Start menu can be accessed by clicking the Windows icon located at the bottom left of the taskbar. If we consider the Windows Desktop to be an airport terminal, the Start menu is the departure board or information desk. It is the area where we see what is available: apps, files, folders, settings, and power options. You can think of it as a quick-access menu.

Built-in Tools and Apps

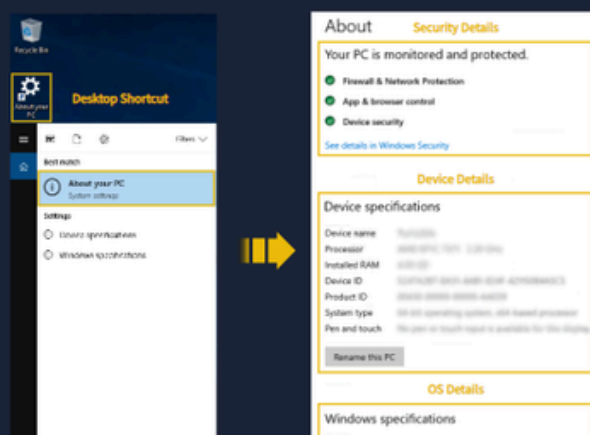
Windows ships with many useful built-in tools and applications that you will use daily. Beyond the wide range of settings available to manage your system, Windows also includes simple but powerful tools such as **Notepad** for editing text files and **File Explorer** for navigating and managing files. These tools are available immediately and form the foundation of everyday Windows usage. All of them can be accessed via the Start menu and search bar. The built-in tools on Windows are like airport services and amenities, the things already available in the terminal to help you get tasks done.



Getting System Information

Windows includes a built-in **Settings** application that allows you to configure and view information about your system. To get to know your new work PC better, we'll start by investigating the machine we're working with. Within the Settings app, there's a helpful section called **About your PC**, that provides key details about the system. A shortcut to **About your PC** has been created on the Desktop. Open it now. Alternatively, you can practice using the **Start Menu** or **Search** feature to locate it.

The **About your PC** page provides an overview of security, device, and operating system information. We'll use this information to become familiar with our environment and begin answering the first questions in this task.



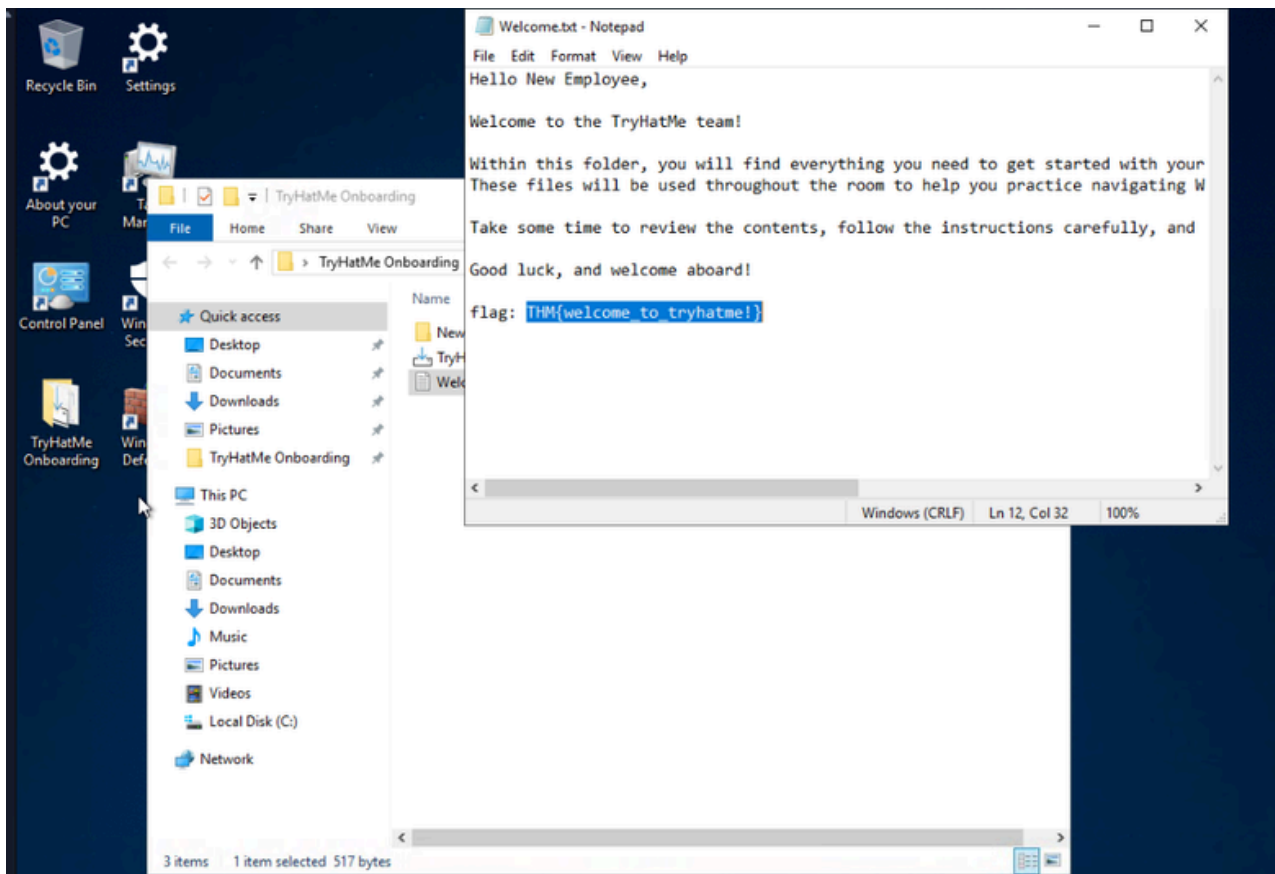
File Exploration and Management

Great, now we have an understanding of the system we are working with. Let's take a look at how Windows handles file exploration and management as we access some of the onboarding documents for your new position. Before we dive into using **File Explorer**, let's quickly cover how Windows organizes files and directories. Windows uses a hierarchical folder structure, meaning folders can contain other folders and files inside them. This structure helps keep data organized and makes it easier to locate information as systems grow larger. Common locations, such as the Desktop, Documents, and Downloads, serve as primary directories for storing files. Within these locations, subfolders are used to group related files together. Understanding this layout will make navigating the system and managing files much easier.

If we continue the airport analogy, File Explorer acts as the directory and floor map of the terminal, helping you navigate between different areas and locate what you need.

You can now open the **TryHatMe Onboarding** folder on the Desktop. Alternatively, you can open the File Explorer application, which is pinned to the Taskbar, or use the Start menu to locate the folder.

1. Open the **TryHatMe Onboarding** folder on your Desktop
2. This opens the folder in the Windows File Explorer
3. Options for viewing, sharing, and editing folders and files
4. Clicking the folder name will show us the full path to the chosen folder
C:\Users\Administrator\Desktop\TryHatMe Onboarding
5. The displayed contents of our chosen folder
6. We can use the File Explorer's built-in search function to search our folder



Answer the questions below

Please ensure the lab machine is open in split-screen, then take a look at the computer's Desktop.

After opening **About your PC**, navigate to the **Device specifications** section.

What is the **Device name** specified?

✓ Correct Answer

Continue looking through the **Device specifications**.

How much RAM is installed on your new work PC?

✓ Correct Answer

Scroll down to the **Windows specifications** section.

Which **Version** of Windows Server 2019 Datacenter is installed?

✓ Correct Answer

Explore the **TryHatMe Onboarding** folder located on your computer's Desktop.

What is the flag value found within **Welcome.txt**?

✓ Correct Answer

Applications are the programs and tools you use to perform tasks on your computer, from browsing the web and editing photos to managing settings on your PC. We already discussed a few of the built-in applications Windows provides, but knowing how to install, update, and remove applications is a core skill for everyday Windows use. Let's briefly revisit the airport analogy to gain a better understanding of this concept. Updating your OS or apps is like changing flights or reserving a seat. Installing a new app is comparable to scheduling a new flight, and removing apps is like canceling a booking you no longer need. These three processes enable you to make the necessary changes to your system, ensuring you have exactly what you need and that your system remains secure.

Updating Your Applications

Keeping your operating system and applications up to date is an important part of maintaining a secure and stable system. Updates will often include security patches, performance updates, and bug fixes.

Windows Updates

Windows includes a built-in update tool called Windows Update, which keeps the OS and some native applications and security features up to date. Windows Update can be accessed through the Settings app and may install updates automatically, depending on your configuration.

Updating Applications

Application updates work differently depending on how the software is installed.

- Built-in applications may update automatically in the background
- Third-party applications often include their own update mechanisms
- Some applications will prompt you to update upon launch
- Some require you to check for updates or download a new installer manually

Installing Applications

Now that you've seen how updates work within the Windows OS and for applications, let's take a look at installing new ones.

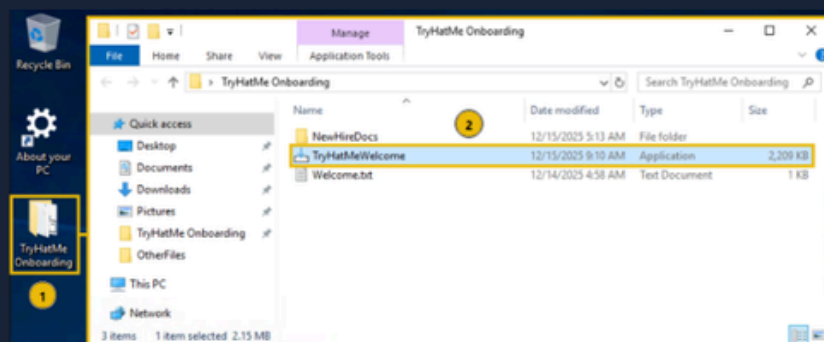
1. **Microsoft Store:** Provides a curated and safe option for installing apps to Windows, although it is not available by default on Windows Server
2. **From the Internet:** In many environments, apps are installed by downloading an installer directly from a trusted vendor's website. They usually come in an `.exe` or `.msi` file and guide the user through the installation process

Hands-On Installation

Great, now you have a solid understanding of updating and installing new applications within a Windows environment. Let's get hands-on and install an app!

1. Within the **TryHatMe Onboarding** folder on your Desktop, you'll find an installer named **TryHatMeWelcome**
2. Proceed by double-clicking the installer to start the installation process, then answer the first question of the task

Note: Feel free to click through the installation options; remember that this is a virtual environment, safe for testing and experimentation.



Uninstalling Applications

In a Windows environment, there are multiple ways to uninstall programs.

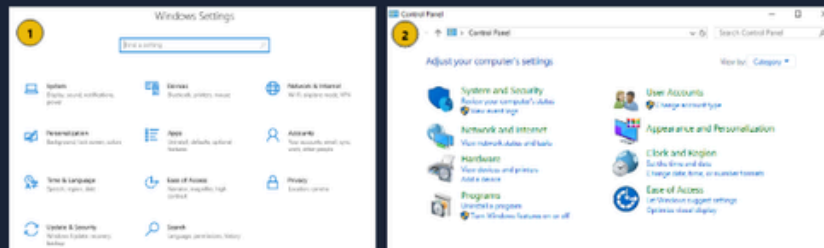
- Using the Microsoft Store for installed applications
- Add or remove programs feature in system settings
- Uninstall a program section of the Control Panel
- Using an application's built-in uninstaller

Diving Into Settings

In the previous task, you explored the **About your PC** section of Windows settings. Now, we will take a closer look at the abundance of configurations available to us. There are two primary ways in which a Windows user can modify their environment. There are existing shortcuts for each of the tools below placed on your Desktop.

1. **Windows Settings:** A modern, centralized location for configuring system, device, personalization, and security settings in Windows
2. **Control Panel:** A legacy management interface that provides access to older system configuration tools still required for specific administrative tasks

Windows Settings and the **Control Panel** enable you to view and modify how your Windows system operates. From these two applications, you can manage system preferences, including display and audio settings, user accounts, apps, network options, accessibility features, and security configurations.



The Task Manager

Task Manager is a built-in Windows tool that allows you to monitor what is happening on your system in real time. It allows you to view running applications and background processes, as well as check system performance, including CPU and memory usage. A shortcut to Task Manager has been placed on your workstation's Desktop. Go ahead and open it now to view your currently running applications and check on your system's performance.

Task Manager has five tabs to help you keep track of your system.

1. **Processes:** Currently running apps and background processes, and their resource usage
2. **Performance:** Graphs and statistics for system resources such as CPU, memory, and network
3. **Users:** Currently logged-in users and used resources
4. **Details:** A more technical view of running processes, including process IDs (PIDs)
5. **Services:** Windows services and their current status (running or stopped)

Native Windows Security

Windows offers built-in security tools designed to help protect your system from threats such as malware, insecure applications, and unauthorized network access. These are enabled by default and allow the monitoring and control of your system's security.

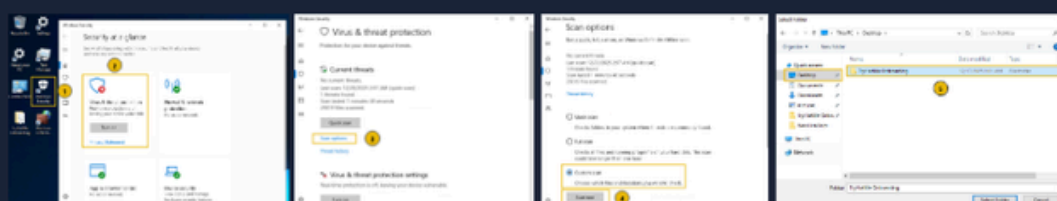
Windows Security

The **Windows Security** application is your central dashboard for managing Windows' built-in protection measures. It is divided into four main sections, each focusing on a different area of system security.

- **Virus & threat protection:** Helps detect and remove malicious software using real-time protection and customizable scans
- **Firewall & network protection:** Controls incoming and outgoing network traffic to help prevent unauthorized access
- **App & browser control:** Protects users from potentially unsafe apps, files, and websites
- **Device security:** Provides hardware-based protections that help secure the system

Let's take a closer look at the **Virus & threat protection** section of Windows Security on your workstation. Go ahead and open it up using the Desktop shortcut or via the Start menu now. Next, we will run a custom scan of your onboarding folder to ensure that there are no malicious files present.

1. Open **Windows Security** using the shortcut on your Desktop
2. Select the **Virus & Threat protection** section
3. Choose **Scan options**
4. Select **Custom scan**, then **Scan now**
5. Select the **TryHatMe Onboarding** folder on your Desktop as the target of your scan

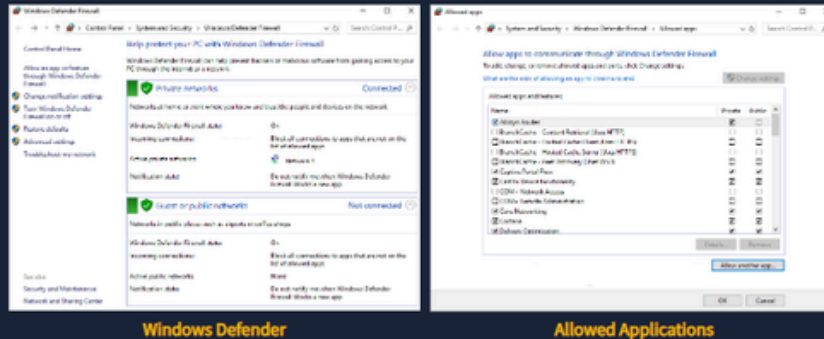


Your custom scan will have found a completely safe test file, and this process shows how Windows is capable of scanning for and detecting threats. Keep in mind that this is just one aspect of the security measures Windows has in place to help you secure your system. Go ahead and click **See details** to investigate the file and answer the final question of the task.

Windows Defender Firewall

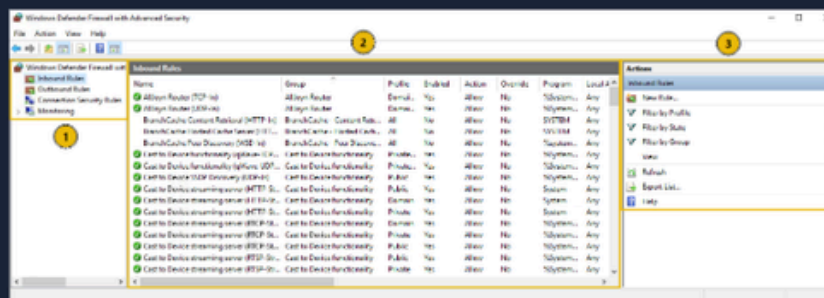
Windows Defender Firewall is a built-in firewall designed to help protect your computer from unauthorized network traffic. It monitors network connections and applies rules that determine whether the connections are allowed or denied. The firewall operates on different network profiles, allowing you to create custom rules or specify applications that are permitted.

- **Domain:** Used when a system is connected to an organization's domain network
- **Private:** Intended for trusted networks, such as a home or lab environment
- **Public:** Used for untrusted networks, such as public Wi-Fi



Checking out the Advanced settings of Windows Defender Firewall, you can view

1. An overview of your firewall's inbound, outbound, and connection rules
2. A detailed view of each rule, including name, group, network profile, status, and action
3. Create new rules or filter your current view



Answer the questions below

Use the **TryHatMeWelcome** installer located within the **TryHatMe Onboarding** folder.
What is the flag value you receive after installing and running the application?

THM{your_first_day!}

✓ Correct Answer

Investigate the **Time & Language** section of the **Windows Settings** app.
Which country or region is your computer currently set to?

United States

✓ Correct Answer

Open the **Task Manager** on your workstation's Desktop and navigate to the **Users** tab.
Which account is currently logged in?

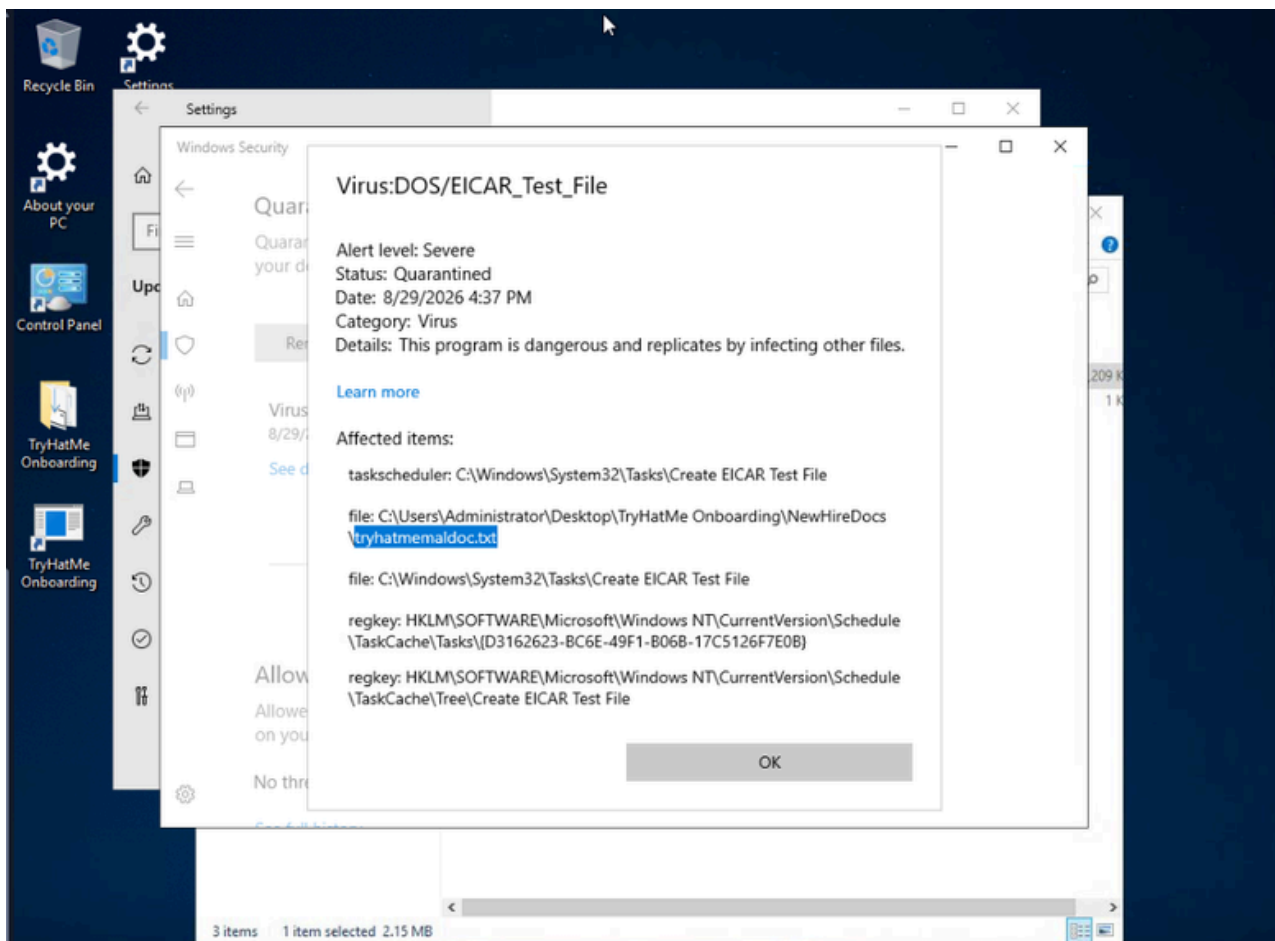
Administrator

✓ Correct Answer

After performing your custom scan, click **Virus:DOS/EICAR_Test_File** and select **See details**.
What is the file name shown in the **Affected items** section?

tryhatmemaldoc.txt

✓ Correct Answer



Task 4 Conclusion

Nicely done! You've reached the end of **Windows Basics**. In this room, you explored the fundamentals of the Windows operating system. You navigated the Windows interface, examined system properties, learned about application management, worked with files and folders, and managed security settings that make up a Windows environment.

Through a hands-on lab, you completed your first day at **TryHatMe**, using the **Windows Server 2019** operating system to read files, install programs, and run security scans on your new workstation. These skills form the foundation for both general system usage and deeper technical understanding later on in your learning and professional journey.

Key Terminology

Let's recap the core terminology and applications you've learned about in this room. These definitions will help solidify your understanding before moving on to further learning.

- **Desktop:** The main workspace where files, folders, and shortcuts live
- **Taskbar:** A control strip that provides access to applications, tools, settings, and notifications
- **Start Menu:** The primary way to access applications, settings, and power options, signified by the Windows logo
- **Search:** A quick access method of locating applications, settings, and files by entering search terms
- **File Explorer:** The built-in Windows tool to browse, manage, and organize files and folders
- **Windows Update:** A built-in update tool that helps keep your OS, native apps, and security features up to date
- **Microsoft Store:** The native Windows application for installing trusted applications
- **Windows Settings:** A centralized location for configuring system, device, personalization, and security settings
- **Control Panel:** The legacy management interface that provides access to system configuration options
- **Task Manager:** A Windows tool for monitoring what is happening on your system in real time
- **Windows Security:** The central dashboard for managing Windows built-in security tools
- **Windows Defender Firewall:** The firewall designed to help protect your system from unauthorized network traffic

Further Learning

In the following rooms of this module, you'll use the command-line (CLI), a text-based interface for interacting with the operating system, to get to know the **Linux** OS and further explore Windows.

- [Linux CLI Basics](#)
- [Windows CLI Basics](#)

Answer the questions below

Complete the room and continue on your cyber learning journey!

No answer needed

✓ Correct Answer

Great work Abhishek26! Room completed!

Your skills are skyrocketing!

Completed tasks

✓ 4

Points earned

🎯 64

Streak

🔥 9



Dashboard

Learn

Practice

Compete



20

10



Pre Security > Operating Systems Basics > Linux CLI Basics



Linux CLI Basics

Premium room

Get comfortable navigating through the Linux CLI.

🕒 45 min 👤 71,671 🏆

Save Room

476 Recommend

Connectivity

Options

Room progress: 12%

Task 1 Introduction

In cyber security, Linux is everywhere, powering servers, security tools, and even hacking environments. However, before you can defend systems or investigate incidents, you need to know how to navigate the Linux OS using the **command-line interface (CLI)**. This room takes you on a guided mission as a brand-new intern in a cyber security team. We will explore the absolute basics of Linux CLI through hands-on tasks, simple missions, and beginner-friendly explanations.

By the end of this task, you'll understand what the terminal is, why it matters, and how you'll use it throughout the room.

Learning Objectives

- Understand what the Linux terminal is and what it's used for.
- Feel comfortable interacting with the Linux environment.
- Navigate through the Linux filesystem with basic commands.

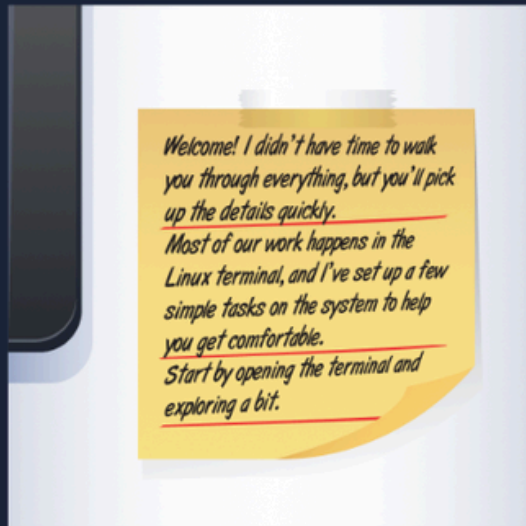
Prerequisites

- Operating Systems: Introduction
- Windows Basics

Storyline: Your First Day on the Job

Congratulations, today is your first day as an IT Support Engineer with the Cyber Operations Support Team. You were supposed to get a proper introduction and a tour of the tools you'll be using, but things in this field don't always go as planned.

Your supervisor left in a hurry to deal with an urgent incident. Before they ran out the door, they left you a short note on your desk:



So here you are on your own for now, faced with a terminal prompt and a bit of mystery. Don't worry, though. Each step in this room will guide you through exactly what to do. You'll learn how to move around the system, read files, and gather information - the same basic skills that cyber security professionals use every day.

Ready to begin?

Answer the questions below

What does "CLI" stand for?

command-line interface

✓ Correct Answer

Task 2 ✓ Navigation Mission: "Find the Missing Notes"



Before starting, click on the **Start Lab Machine** button below. This will start your lab machine in split view. If split view is not visible, press the **Show Split View** button at the top of the page.

Your virtual environment has been set up ✓

All machine details can be found at the top of the page.



Lab machine ⓘ

Status: ● On

🔗 See connection methods

A Quick Note About the Terminal

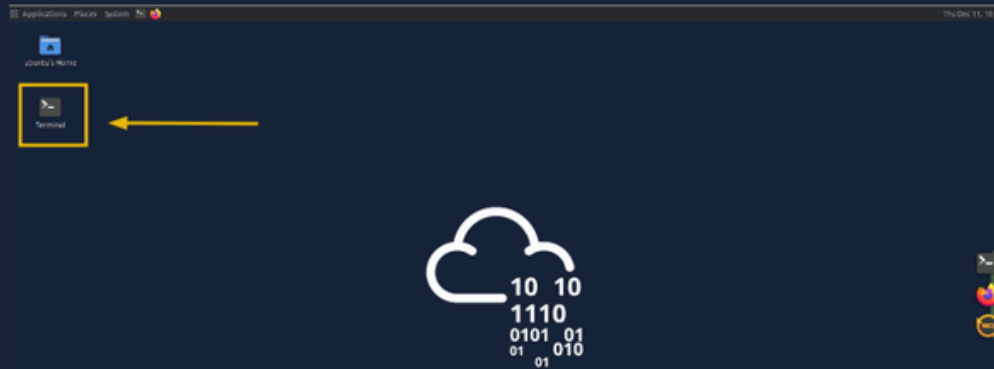
Before we begin exploring, let's take a moment to familiarize ourselves with the tool we're using.

The **terminal** is a text-based interface for controlling a Linux system. Instead of interacting with the graphical interface, you type commands that tell the computer exactly what to do. Cyber security professionals use it because:

- It's faster than clicking around
- It gives more control
- Many security tools only run in the terminal

If the terminal feels unfamiliar at the moment, that's completely normal.

By the end of this room, it will feel far more comfortable, and you'll see it as a straightforward tool rather than something intimidating. Double-click on the **Terminal** file present on the Desktop, as shown below:



Interacting With the Terminal

Now that you know what the terminal is, it's time to start using it.

Your supervisor mentioned leaving a few tasks in the document `mission_brief.txt`, somewhere on the system; which means your first mission is to find them. To do that, you'll need to learn how to move around the Linux filesystem.

This task will introduce the core navigation commands that every Linux user learns on their first day.

Step 1: "Where Am I?"

When you first open a terminal, you might not know what part of the system you're in.

Use this command to find out: `pwd`

```
ubuntu@tryhackme:~$ pwd
/home/ubuntu
```

It stands for "print working directory", which basically means "show me the folder I'm currently in".

Step 2: "What's Around Me?"

Now that you know where you are, let's see what files and folders are here: `ls`

```
ubuntu@tryhackme:~$ ls
Desktop  Downloads  Pictures  Templates  logs
Documents Music      Public    Videos    projects
```

This lists the content of the current directory. If we need more details, we can try: `ls -l`

```
ubuntu
@tryhackme:~$ ls -l
total 44
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Desktop
drwxr-xr-x 6 ubuntu ubuntu 4096 Dec 11 12:45 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 16 2024 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Feb 27 2022 Music
```

The output displays important information about the files and directories like file sizes, permissions, dates, and more.

Hidden Files

In order to get the hidden files in the directory, we can append the command to `ls -al`, and it will display all the hidden files present in the directory, as shown below:

```
ls -al Command

ubuntu@tryhackme:~$ ls -al
total 144
drwxr-xr-x 24 ubuntu ubuntu 4096 Feb 10 10:48 .
drwxr-xr-x  3 root    root    4096 Feb 10 10:36 ..
-rw-----  1 ubuntu ubuntu  439 Feb 10 06:47 .Xauthority
-rw-rw-r--  1 ubuntu ubuntu    0 Sep 12 2024 .Xresources
-rw-r--r--  1 ubuntu ubuntu  111 Oct  3 2024 .appport-ignore.xml
-----
drwxr-xr-x  2 ubuntu ubuntu 4096 Feb 27 2022 Templates
drwxr-xr-x  2 ubuntu ubuntu 4096 Feb 27 2022 Videos
drwxr-xr-x  4 ubuntu ubuntu 4096 Dec 11 12:29 logs
drwxr-xr-x  5 ubuntu ubuntu 4096 Dec 11 12:29 projects
drwx-----  3 ubuntu ubuntu 4096 Sep 12 2024 snap
```

Hidden files aren't really `secret`; they start with a dot `.`, and Linux hides such files by default.

Step 3: Let's Move Around

To walk through the filesystem, we can use: `cd <directory>`. For example: `cd Documents`, and this will change our directory to `Documents`, as shown below

```
cd Command

ubuntu@tryhackme:~$ cd Documents/
ubuntu@tryhackme:~/Documents$ pwd
/home/ubuntu/Documents
```

To go "back" one level, we will use the command `cd ..`, as shown below:

```
cd Command

ubuntu@tryhackme:~/Documents$ cd ..
ubuntu@tryhackme:~$ pwd
/home/ubuntu
```

Step 4: Learn the Power of "Find"

Let's now use one of the handy tools to `find`. As the name suggests, this built-in utility is used to locate files within the file system. Here's a simple version of the command: `find <starting_point> -name <filename>`. Since your supervisor mentioned that the file `mission_brief.txt` resides somewhere in your home directory, begin the home directory symbol: `~`. So we will run the command: `find ~ -name mission_brief.txt`, as shown below:

```
find Command

ubuntu@tryhackme:~$ find ~ -name mission_brief.txt
/<REDACTED-PATH>/mission_brief.txt
```

Please note that this may take a moment, as Linux will check every folder inside your home directory. If the file exists, Linux will print the `full path` to it. The above result shows that the command has successfully located the file and provided us with its complete path, allowing us to navigate and read the file's content.

Let's use the `cd` command to navigate to the folder. We can also do `ls` to confirm the presence of the file, as shown below:

```
cd Command

ubuntu@tryhackme:~$ cd /<REDACTED-PATH>/
ubuntu@tryhackme:~/<REDACTED-PATH>$ ls
mission_brief.txt
```

Step 5: Read the File

Another useful utility is `cat`. This is used to read the content of the file. To read this file, we will run the command `cat mission_brief.txt` to get the content, as shown below:

```
cat Command

ubuntu@tryhackme:~/<REDACTED-PATH>$ cat mission_brief.txt
Great job finding your way around the terminal.

Your next assignment is to collect a small system report:
- Who you're logged in as
- The kernel version
- Total disk space
- The name of this Linux distribution

Once you gather those details, you'll be ready for the next step.

FLAG: <REDACTED>
```

Perfect. This is great progress. You have completed the task.

After answering the questions, let's proceed to the next task and learn how to retrieve system information using the commands in the [CLI](#).

Answer the questions below

What is the full path of the `mission_brief.txt` file found on the system using the `find` command?

/home/ubuntu/Documents/.research/archive/mission_brief.txt

✓ Correct Answer

What is the flag hidden inside the `mission_brief.txt` file?

MISSION-FOUND

✓ Correct Answer

Now your supervisor wants you to gather some basic information about the system you're working on. This kind of information helps cyber security teams understand what environment they're operating in, what version of Linux they're using, and what resources are available.

Think of this as taking a quick health check of the machine.



Your Next Assignment

When you opened `mission_brief.txt`, it contained a new message:

Great job finding your way around the terminal. Your next assignment is to collect a small system report:

- Who you're logged in as
- The kernel version
- Total disk space
- The name of this Linux distribution

You're still on your own for now, but at least the instructions are clear: collect system info and send it back. Let's break the job into small steps.

Step 1: "Who Are You Logged in As?"

The simplest command in Linux also happens to be one of the most useful: `whoami`

```
whoami Command

ubuntu@tryhackme:~$ whoami
ubuntu
```

This prints your current username.

Step 2: "What System Are You On?"

To see details about the operating system, kernel version, and architecture, use: `uname -a`

```
uname -a Command

ubuntu@tryhackme:~$ uname -a
Linux tryhackme <REDACTED>-aws #17-Ubuntu SMP Mon Sep  2 13:48:07 UTC 2024 x86_64 x86_64 x86_64 GNU/Linux
```

This gives a full line of system information - great for understanding exactly what environment you're working in.

Breakdown of the Information

- **Linux:** The system is running the Linux kernel.
- **tryhackme:** The hostname (the computer's name).
- **<REDACTED>-aws:** The kernel version installed on the machine.
- **x86_64:** The hardware platform (also 64-bit).
- **GNU/Linux:** The operating system type (Linux kernel + GNU tools).

If you only want the operating system name, you can try: `uname`

```
uname Command

ubuntu@tryhackme:~$ uname
Linux
```

But for now, `uname -a` gives the fuller picture.

Step 3: Check Disk and Storage Info

In real-world environments, you'll often need to check disk usage or available space, especially before running tools or analyzing logs. A simple command for readable output is: `df -h`

```
df -h Command

ubuntu@tryhackme:~$ df -h

Filesystem      Size  Used Avail Use% Mounted on
/dev/root        --G   12G   --G   17% /
tmpfs            1.9G   0    1.9G   0% /dev/shm
tmpfs            774M   1.2M  773M   1% /run
tmpfs            5.0M   0    5.0M   0% /run/lock
tmpfs            387M   192K  387M   1% /run/user/1000
tmpfs            387M   172K  387M   1% /run/user/114
```

The `-h` means "human readable"; it shows sizes like 2G or 500M instead of long bytes-only numbers.

Breakdown of the Information

- `/dev/root` is the main disk of the system with `--G` total, `12G` used, `<REDACTED>G` free, and is `17%` full.
- `tmpfs` entries are temporary filesystems stored in RAM, not on the physical disk.
- `/dev/shm` is a shared memory area with `1.9G` available and `0` used.
- `/run/user/114` is similar temporary storage for another system user, also `387M` total and mostly empty.

Step 4: Read a System File

Linux stores configuration and informational files in the `/etc` directory.

To practice navigating and reading files, head into `/etc` by running `cd /etc` and then list what's inside: `ls`

```
cd Command

ubuntu@tryhackme:~$ cd /etc
ubuntu@tryhackme:/etc$ ls
ImageMagick-6      cloud              firefox            hp                 logcheck
ModemManager       compizconfig      fonts              ifplugd            login.defs
NetworkManager     console-setup     fstab              init               logrotate.
---
-----
chatscripts        ethertypes        hosts.deny         localtime          openvpn
ubuntu@tryhackme:/etc$
```

Let's now use `cat` to read the `os-release` file that almost every Linux system contains: `cat os-release`, as shown below:

```
cat Command

ubuntu@tryhackme:/etc$ cat os-release
PRETTY_NAME="Ubuntu 24.04.1 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.1 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
```



This file provides details about the Linux distribution that are often clearer than those provided by the `uname` command.

Great work. With just a few commands, we were able to retrieve the system information as requested by the supervisor.

Wrapping Up Your First Day

You've navigated the filesystem, searched for hidden files, and gathered system information, solid progress for your first steps into Linux. Before you finish your day, your supervisor has left one final task to test everything you've learned so far.

Nothing too hard, just a quick check to make sure you can put the basics together on your own.

Mini Challenge

You only know the name of the file: **day1_report.txt**

Your job:

1. Use the `find` command to locate it somewhere in your home directory.
2. Navigate to the folder where it was found.
3. Read the file contents using `cat`.
4. Use it for the the last question in this task.

No step-by-step instructions this time. You've earned the confidence to do this on your own.

Answer the questions below

What is the username returned by the `whoami` command?

✓ Correct Answer

What is the kernel version shown by `uname -a` ?

✓ Correct Answer

How much free disk space does `df -h` report?

✓ Correct Answer

What is the message written inside **day1_report.txt**?

✓ Correct Answer

Task 4 Conclusion

And that's your first day on the Cyber Operations Support Team.

You learned how to:

- Navigate the Linux filesystem
- Search for files
- Inspect system information
- Read important configuration files
- Follow clues and complete missions inside a Linux environment

These skills may seem simple, but they're the building blocks for everything that comes later: file permissions, users & groups, processes, package management, and eventually real security tooling.

You're off to a strong start.

Answer the questions below

Continue to complete the room.

✓ Correct Answer

Great work Abhishek26! Room completed!

Your skills are skyrocketing!

Completed tasks

4

Points earned

56

Streak

10

Pre Security > Operating Systems Basics > Windows CLI Basics

Windows CLI Basics Premium room

Explore what Windows CLI is, how to navigate, and interact with the system using Windows CLI.

60 min 65,540

Save Room

340 Recommend

Connectivity

Options

Room progress: 16%

Active machines information

IP Address: Shown in 0min 53s

Task 1 Introduction

In the previous [Linux CLI Basics](#) room, you took your first steps into the Linux command line. Today, you're continuing your journey, but this time on **Windows**. Windows is one of the most widely used operating systems in workplaces worldwide. As someone interested in cyber security, you will often investigate or troubleshoot **Windows machines**, especially user desktops and laptops.

In this room, you'll learn how to use the **Windows Command Prompt** to navigate files, search for information, and gather basic system details using the same hands-on, story-driven approach as before.

From a security perspective, Windows is important because many real-world attacks and investigations involve Windows systems.

What Is the Windows Command Line?

The **Command Prompt** (often referred to as CMD) is a text-based interface for interacting with the Windows operating system. Instead of clicking folders and menus, you type commands to tell the system exactly what you want to do, such as listing files, moving between folders, or checking system information. It might look simple, but it's a powerful and widely used tool.

Learning Objectives

By the end of this room, you will be able to:

- Use the Windows command line confidently
- Navigate folders without clicking
- Find files when you only know their name
- Read files using the terminal
- Collect basic system and network information

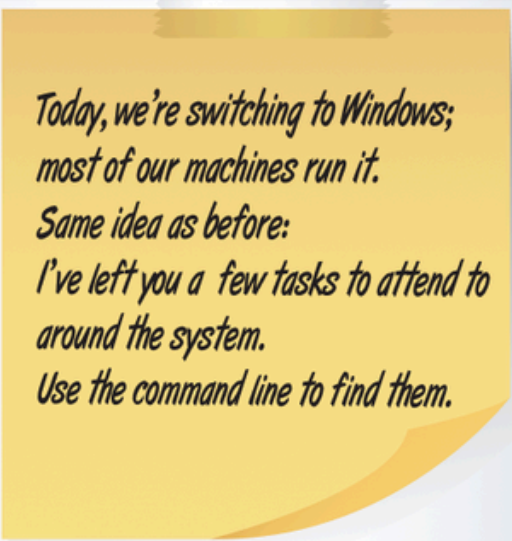
Prerequisites

This room expects you to have explored the following rooms to understand better:

- [Operating Systems](#)
- [Linux CLI Basics](#)

Storyline: Day 2 on the Job

You arrive for **the second day** of your internship. Your supervisor greets you with a quick update:



*Today, we're switching to Windows;
most of our machines run it.
Same idea as before:
I've left you a few tasks to attend to
around the system.
Use the command line to find them.*

So here you are on your own for now, faced with a terminal prompt and a bit of mystery. Don't worry, though. Each step in this room will guide you through exactly what to do. You'll learn how to move around the system, read files, and gather information - the same basic skills that cyber security professionals use every day.

Ready to begin?

[Answer the questions below](#)

Continue to the next task.

No answer needed

✓ Correct Answer

Before starting, click on the green button **Start Lab Machine**, shown below:

Your virtual environment has been set up

All machine details can be found at the top of the page.



Lab machine 

Status:  On

 [See connection methods](#)

A Quick Recap About the Terminal

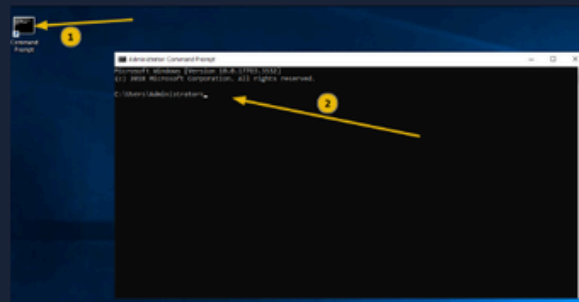
Before we begin exploring, let's take a moment to familiarize ourselves with the tool we're using.



The **terminal** is a text-based interface for interacting with the Windows OS. Instead of clicking windows and folders, you type commands that tell the computer exactly what to do. cyber security professionals use it because:

- It's faster than clicking around
- It gives more control
- Many security tools only run in the terminal

By the end of this room, it will feel far more comfortable, and you'll see it as a straightforward tool rather than something intimidating. Click on the Terminal file present on the Desktop:



Overview

Before you can explore anything on a Windows system, you need to know how to move around and look for files using the command line. In this task, you'll learn how to explore folders, move between directories, search for a file when you don't know where it is, and read the contents of a text file, all using **Command Prompt**.

These are the same kinds of skills you'll use later when troubleshooting systems.

Task: Find the mission_brief.txt file

Your boss has left you another note, with a task to find the task_brief.txt file:

Note: I've left a file called `task_brief.txt` somewhere in your user folder.
You'll need to find it using the command line and read what it says.

"I've left a file called **`task_brief.txt`** somewhere in your user folder.
You'll need to find it using the command line and read what it says."

You're only given the **name of the file**, not its location.

Let's explore how to navigate through the Windows filesystem using the command line and how to locate a file without knowing its actual location.

Step 1: Where Am I?

Before doing anything else, open the terminal on the Desktop, and check your current location by typing the command `cd`, as shown below:

```
C:\Users\Administrator>cd
C:\Users\Administrator
```

This shows the full path of the directory you're currently in. On Windows, this is usually your **user folder**. Please note that the `cd` command is also used to change the directory, which we will use later in the task.

Step 2: What's Around Me?

Now, list the contents of the current directory using the command: `dir`

This command will list down the files and the folders present in the current directory, as shown below:

```
C:\Users\Administrator>dir
Volume in drive C has no label.
Volume Serial Number is A8A4-C362

Directory of C:\Users\Administrator

01/04/2026  10:14 PM  <DIR>          .
01/04/2026  10:14 PM  <DIR>          ..
03/04/2024  10:17 AM  <DIR>          .ms-ad
01/04/2026  10:14 PM  <DIR>          .research
10/14/2022  10:06 AM  <DIR>          3D Objects
10/14/2022  10:06 AM  <DIR>          Contacts
01/04/2026  10:23 PM  <DIR>          Desktop
01/04/2026  10:14 PM  <DIR>          Documents
01/04/2026  10:14 PM  <DIR>          Downloads
10/14/2022  10:06 AM  <DIR>          Favorites
10/14/2022  10:06 AM  <DIR>          Links
10/14/2022  10:06 AM  <DIR>          Music
10/14/2022  10:06 AM  <DIR>          Pictures
10/14/2022  10:06 AM  <DIR>          Saved Games
02/26/2024  01:49 AM  <DIR>          Searches
10/14/2022  10:06 AM  <DIR>          Videos
             0 File(s)              0 bytes
            16 Dir(s)  3,022,721,024 bytes free
```

You'll see files and folders that are visible by default. Take a moment to look around; not everything you need will always be obvious. From the output, we can see that the command has returned 16 directories.

Step 3: Are There Hidden Files?

Some files and folders on Windows are marked as hidden, which means they don't appear in a normal listing. To show everything, including hidden items, run: `dir /a`

The output clearly indicates that more than 12 hidden folders were found. It is important to note that hidden doesn't mean secret; it just means Windows hides them by default.

Let's use the `cd` command to navigate through the folders. We can use the format `cd folder_name` to move to the specified folder. The command `cd Documents` will move us to the Documents folder, as shown below. To move back one level, we can use the `cd` command.

Perfect.

You have not only learned how to navigate Windows directories using the CLI, but also how to find files and read their contents.

The content of `task_brief.txt` leads us to our next task: inquiring about the system we are currently using. Let's explore that in the next task.

Answer the questions below

What is the full path of the `task_brief.txt` found on the system?

`C:\Users\Administrator\Documents\Notes\research_yn6\exports_imv\screenshots\notes_wi6`

✓ Correct Answer

What message and flag are written inside `task_brief.txt`?

`TASK-BRIEF-FOUND`

✓ Correct Answer



```
Administrator: Command Prompt
Directory of C:\Users\Administrator\Documents\Notes\research_yn6\exports_imv\screenshots\notes_wi6
01/04/2026  10:14 PM                220 task_brief.txt
              1 File(s)                220 bytes

Total Files Listed:
              1 File(s)                220 bytes
              0 Dir(s) 44,705,247,232 bytes free

C:\Users\Administrator>cd Documents\Notes\research_yn6\exports_imv\screenshots\notes_wi6
C:\Users\Administrator\Documents\Notes\research_yn6\exports_imv\screenshots\notes_wi6>dir
Volume in drive C has no label.
Volume Serial Number is A8A4-C362

Directory of C:\Users\Administrator\Documents\Notes\research_yn6\exports_imv\screenshots\notes_wi6
01/04/2026  10:14 PM                220 task_brief.txt
              1 File(s)                220 bytes
              0 Dir(s) 44,696,342,528 bytes free

C:\Users\Administrator\Documents\Notes\research_yn6\exports_imv\screenshots\notes_wi6>type task_brief.txt
Nice work getting into the Windows command line.

Your next tasks:
- Identify who you are logged in as
- Check the computer name
- Find the Windows version
- Check the machines IP address
```

Task 3 Gathering System Information on Windows

Overview

Now that you know how to move around and find files using the Windows command line, it's time to learn how to **ask the system questions about itself**. cyber security and IT professionals do this all the time. Before fixing a problem or investigating an incident, they first want to know:

- Who am I logged in as?
- What machine is this?
- What version of Windows is it running?
- How is it connected to the network?

In this task, you'll learn how to gather that information step by step.



Step 1: Who Am I Logged In As?

When working on a system, one of the first things to check is **which user account you're using**. This matters because different users can have different permissions.

Run the following command: `whoami`. This command prints the username of the account you're currently logged into.

```
C:\Users\Administrator>whoami
Administrator
```

Step 2: What Is the Name of This Computer?

Every Windows machine has a name. In workplaces, this helps identify network systems. To see the computer's name, run: `hostname`.

```
C:\Users\Administrator>hostname
```

You'll see a short name printed in the terminal.

Step 3: What Version of Windows Is This?

Next, let's look at details about the operating system itself. Run: `systeminfo`

This command prints a lot of information. Don't worry, you're **not expected to understand everything** yet.

```
C:\Users\Administrator>systeminfo

Host Name: [REDACTED]
OS Name: Microsoft Windows Server 2019 Datacenter
OS Version: [REDACTED]
OS Manufacturer: Microsoft Corporation
OS Configuration: Standalone Server
OS Build Type: Multiprocessor Free
Registered Owner: EC2
Registered Organization: Amazon.com
Product ID: 00430-00000-00000-AA750
Original Install Date: 1/17/2021, 2:59:00 PM
System Boot Time: 1/4/2026, 9:37:30 PM
System Manufacturer: Amazon EC2
System Model: c5a.medium
System Type: x64-based PC
Processor(s): 1 Processor(s) Installed.
[01]: AMD64 Family 23 Model 1 Stepping 2 AuthenticAMD ~2200 Mhz
BIOS Version: Amazon EC2 1.0, 10/10/2017
Windows Directory: C:\Windows
System Directory: C:\Windows\System32
Root Device: \Device\HarddiskVolume1
System Locale: en-us;English (United States)
Input Locale: en-us;English (United States)
Time Zones: (UTC) Coordinated Universal Time
Total Physical Memory: 4,048 MB
Available Physical Memory: 2,763 MB
Virtual Memory: Max Size: 4,752 MB
Virtual Memory: Available: 2,430 MB
Virtual Memory: In use: 1,342 MB
Page File Location(s): C:\pagefile.sys
Domain: WORKGROUP
Logon Server: \\THMLAB
Netfix(s): 29 Netfix(s) Installed.
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[02]: KB5017263
[03]: KB5017263
[04]: KB5017263
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[817]: KB5017263
[818]: KB5017263
[819]: KB5017263
[820]: KB5017263

```

```
C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : ec2.internal
    Link-local IPv6 Address . . . . . : fe80::5c86:738b:6946:29e8%4
    IPv4 Address. . . . . : 10.177.63.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.177.63.1
```

This information helps analysts understand how a machine connects to the network.

Summary

In this task, you learned how to ask a Windows system simple but important questions using the command line.

You practiced:

- Identifying **who** you are logged in as
- Finding the **name of the computer**
- Checking **Windows version and system details**
- Viewing **basic network information**

These are some of the first commands analysts and IT staff run when they start working on a Windows machine.

Answer the questions below

What is the computer name shown by hostname?

thmlab

✓ Correct Answer

What Windows version is listed in the systeminfo output?

10.0.17763 N/A Build 17763

✓ Correct Answer

Task 4 Conclusion

Overview

You've reached the end of this room - nice work.

In this room, we learned how to interact with a Windows system using the command line instead of relying on the graphical interface. This task is a short wrap-up to reflect on what you've learned and why it's important.

What You Learned

In this room, you practiced using the **Windows Command Prompt** to:

- Navigate files and folders without clicking
- Locate files even when you didn't know where they were
- Reveal hidden files and directories
- Read file contents directly from the terminal
- Gather basic system and network information

These are foundational skills that apply across many roles in IT and cybersecurity.

Why This Matters

In real-world environments:

- Not everything is visible in the graphical interface
- Files may be hidden or buried deep in the system
- Analysts often need quick answers without clicking through menus
- Command-line tools are faster, more precise, and easier to automate

Being comfortable with the Windows command line gives you more control and better visibility into a system.

Answer the questions below

Continue to complete the room.

No answer needed

✓ Correct Answer



Great work Abhishek26! Room completed!

Your skills are skyrocketing!

Completed tasks

✓ 4

Points earned

🎯 32

Streak

🔥 11

Pre Security > Operating Systems Basics > Operating System Security

Operating System Security 👑 Premium room

This room introduces users to operating system security and demonstrates SSH authentication on Linux.

📶 🤖 ⌚ 60 min 👥 223,712 🗣️

🔌 AttackBox Connecting ▼

🔖 Save Room

👍 3977 Recommend

🌐 Connectivity

⚙️ Options ▼

Room progress: 40%

📖 Active machines information

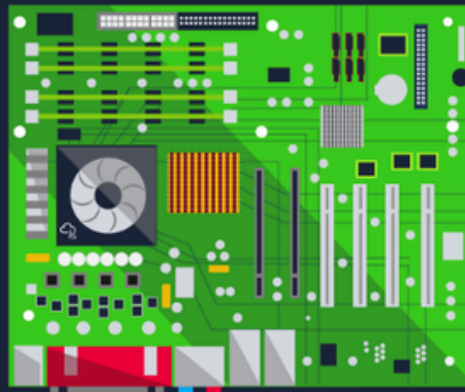
🇺🇸 IP Address: 10.49.65.34

🕒 IP Address: Shown in 0min 26s

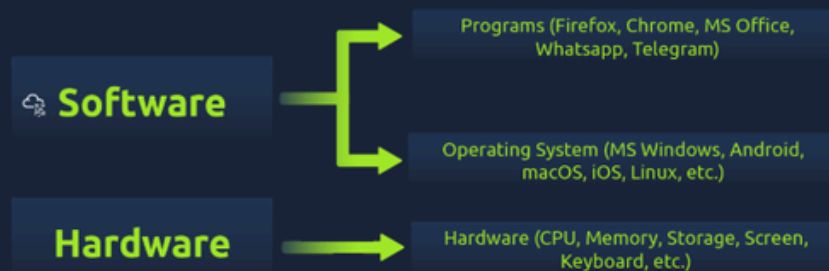


Every day you use a smartphone or a laptop or almost any type of computer, you interact directly or indirectly with an operating system. Operating systems include MS Windows, macOS, iOS, Android, Chrome OS, and Linux. But what is an operating system? To define an operating system, we need to visit one computer term: hardware.

Computer hardware refers to all the computer parts and peripherals that you can touch with your hand. Hardware includes the screen, the keyboard, the printer, the USB flash memory, and the desktop board. As shown in the figure below, the desktop board contains many components, in particular, a central processing unit (CPU) and memory chips (RAM). Although not shown in the image below, the desktop board is usually connected to a storage device (HDD or SSD).



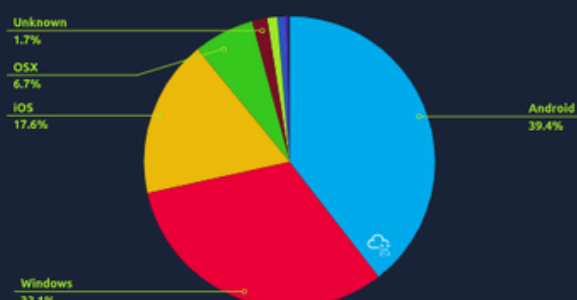
The desktop board is the main part of a computer, and all the other pieces of hardware from keyboard and mouse to screen and printer connect to it. However, hardware components by themselves are useless if you want to run your favorite programs and applications. We need an Operating System to control and “drive” them.



The Operating System (OS) is the layer sitting between the hardware and the applications and programs you are running. Example programs you would use daily might include a web browser, such as Firefox, Safari, and Chrome, and a messaging app, such as Signal, WhatsApp, and Telegram. All the programs and applications cannot run directly on the computer hardware; however, they run on top of the operating system. The operating system allows these programs to access the hardware according to specific rules.

Some operating systems are designed to run on laptops and personal desktops, such as MS Windows 11 and macOS. Other operating systems are designed specifically for smartphones, such as Android and iOS. There are also operating systems intended for servers; examples include MS Windows Server 2022, IBM AIX, and Oracle Solaris. Finally, there are operating systems that you can use on a personal computer and server; one example is Linux. The image below shows the popularity of the different operating systems used to browse the Internet according to Statcounter based on the data collected during January 2022.

Operating System Market Share Worldwide (Jan 2022)



Your smartphone might be running Android or iOS, and you might have plenty of private data on it. Examples include:

1. Private conversations with your family and friends
2. Private photos with family and friends
3. Email client that you use for personal and work communications
4. Passwords saved in the web browser (or even in notes)
5. E-banking apps

The list of confidential and private data goes on. You don't want someone you don't trust to open your phone and go through your photos, conversations, and apps. Hence, you need to secure your phone and its operating system.

The same goes for your laptop or computer running MS Windows, macOS, or Linux. Your computer will most likely contain plenty of information such as:

1. Confidential files related to your work or university
2. Private personal files, such as a copy of your ID or passport
3. Email programs, such as MS Outlook and Mozilla Thunderbird
4. Passwords saved in web browsers and other apps
5. Copy of digital camera and smartphone photos

The list can get very long, depending on the type of user. And considering the nature of the saved data, you want to ensure that your data is secure. When we talk about security, we should think of protecting three things:

- **Confidentiality:** You want to ensure that secret and private files and information are only available to intended persons.
- **Integrity:** It is crucial that no one can tamper with the files stored on your system or while being transferred on the network.
- **Availability:** You want your laptop or smartphone to be available to use anytime you decide to use it.



In the next task, we will discuss common attacks against these security pillars.

Answer the questions below

Which of the following is **not** an operating system?

- AIX
- Android
- Chrome OS
- Solaris
- Thunderbird

Thunderbird

✓ Correct Answer

As we mentioned in the previous task, security is concerned with attacks against:

1. Confidentiality
2. Integrity
3. Availability

In this room, we will focus on three weaknesses targeted by malicious users:

1. Authentication and Weak Passwords
2. Weak File Permissions
3. Malicious Programs

Authentication and Weak Passwords

Authentication is the act of verifying your identity, be it a local or a remote system. Authentication can be achieved via three main ways:

- Something you know, such as a password or a PIN code.
- Something you are, such as a fingerprint.
- Something you have, such as a phone number via which you can receive an SMS message.

Since passwords are the most common form of authentication, they are also the most attacked. Many users tend to use easy-to-guess passwords or the same password on many websites. Moreover, some users rely on personal details such as date of birth and name of their pet, thinking that this is easy to remember and unknown to attackers. However, attackers are aware of this tendency among users.

The National Cyber Security Centre (NCSC) has published a list of the [100,000 most common passwords](#). Let's look at the top 20 passwords in the table below.

Rank	Password
1	123456
2	123456789
3	qwerty
4	password
5	111111
6	12345678
7	abc123
8	1234567
9	password1
10	12345

11	1234567890
12	123123
13	000000
14	iloveyou
15	1234
16	1q2w3e4r5t
17	qwertyuiop
18	123
19	monkey
20	dragon

We can see that 123, 1234, 12345, ..., 123456789, and 1234567890 are on the list. Dictionary words such as password, iloveyou, monkey, and dragon are commonly used. Words not in the dictionary include qwerty, qwertyuiop, and 1q2w3e4r5t; these seemingly complex passwords are very predictable as they follow the keyboard layout.



In brief, if the attacker can guess the password of any of your online accounts, such as your email or social media account, they will be able to gain access to your private data. Therefore, it is vital that you choose complex passwords and use different passwords with different accounts.

Weak File Permissions

Proper security dictates the principle of least privilege. In a work environment, you want any file accessible only by those who need to access it to get work done. On a personal level, if you are planning a trip with family or friends, you might want to share all the files related to the trip plan with those going on that trip; you don't want to share such files publicly. That's the principle of least privilege, or in simpler terms, "who can access what?"

Weak file permissions make it easy for the adversary to attack confidentiality and integrity. They can attack confidentiality as weak permissions allow them to access files they should not be able to access. Moreover, they can attack integrity as they might

modify files that they should not be able to edit.

Access to Malicious Programs

The last example we will consider is the case of malicious programs. Depending on the type of malicious program, it can attack confidentiality, integrity, and availability.



Some types of malicious programs, such as Trojan horses, give the attacker access to your system. Consequently, the attacker would be able to read your files or even modify them.

Some types of malicious programs attack availability. One such example is ransomware. Ransomware is a malicious program that encrypts the user's files. Encryption makes the file(s) unreadable without knowing the encryption password; in other words, the files become gibberish without decryption (reversing the encryption). The attacker offers the user the ability to restore availability, i.e., regain access to their original files: they would give them the encryption password if the user is willing to pay the "ransom."

Answer the questions below

Which of the following is a strong password, in your opinion?

- iloveyou
- 1q2w3e4r5t
- LearnM00r
- qwertyuiop

LearnM00r

✓ Correct Answer



Task 3 Practical Example of OS Security



Your virtual environment has been set up

All machine details can be found at the top of the page.



Attacker machine

Status:  Connected via AttackBox



Lab machine

Status:  On

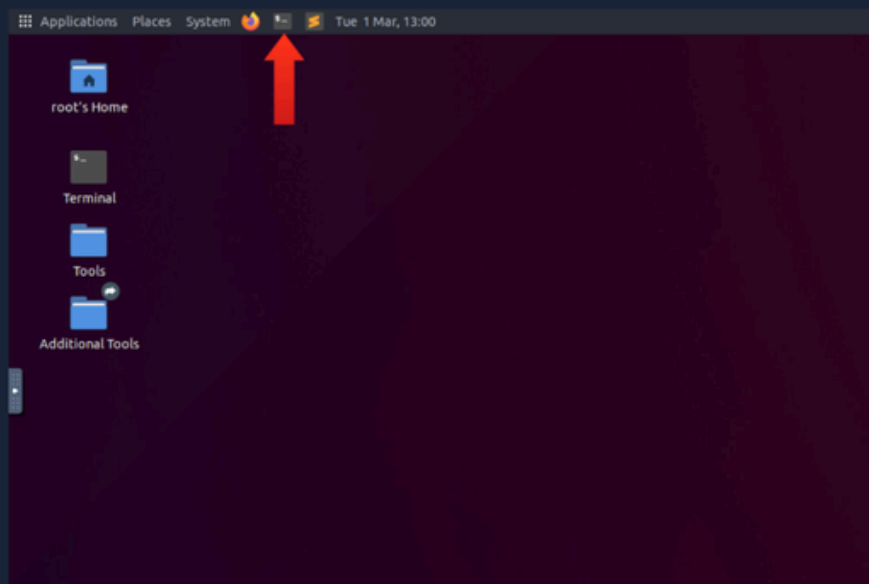
 See connection methods

In one typical attack, the attacker seeks to gain access to a remote system. We can accomplish this attack by tricking the target into running a malicious file or by obtaining a username and a password. We will focus on the latter. After discovering a username, we will try to “guess” the password; furthermore, we will try to escalate our privileges to a system administrator. This account is called **root** on Android, Apple, and Linux systems. While, on MS Windows systems, this account is called **administrator**. The accounts **root** and **administrator** have complete unrestricted access to a system.

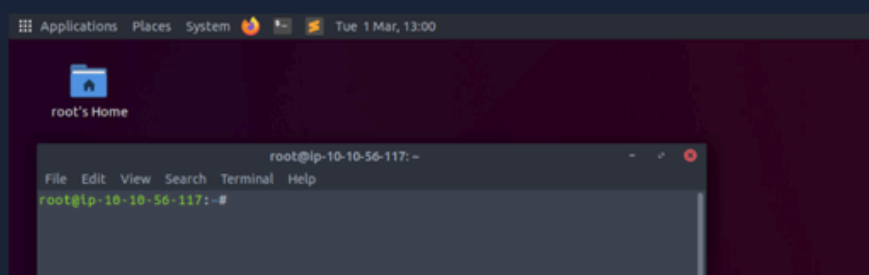


In this task, we will try to hack into a Linux system. We assume that you have never used a Linux system before, and we will explain accordingly.

Start the AttackBox by clicking on the “Start AttackBox” button at the top left of the room. Start the attached machine by clicking on the green “Start Lab Machine” button at the top right of this task. It usually takes a minute or two to load fully. Once they are both ready, you should use the AttackBox, which should be taking the right half of your screen by now.



On the AttackBox, start the terminal by clicking on the terminal icon shown in the image above. You will be writing all the commands you need on the terminal shown below.



We will cover the following Linux commands and explain them throughout this task.

- `whoami`
- `ssh USERNAME@10.49.146.171`
- `ls`
- `cat FILENAME`
- `history`

We were hired to check the security of a certain company. When we visited our client's office, we noticed a sticky note with two words: `sammie` and `dragon` on one of the screens. Let's see if `dragon` is Sammie's password on the lab machine `10.49.146.171`. From the AttackBox's terminal, we will try to log in to Sammie's account by executing `ssh sammie@10.49.146.171`. The remote system will ask you to provide `sammie`'s password, `dragon`.

The first time we connect to a server over SSH, we will get a warning about the server's authenticity and SSH key. We need to answer "Are you sure you want to continue connecting (yes/no)?" with yes.

Please note the following when entering the password over SSH. When you log in via SSH, you won't see that you are typing the password. In other words, **when you are typing the SSH password, you won't see stars, dots, or any indicator on the screen that the password is being typed**. However, the system still receives the password you are entering.

The interaction on the AttackBox's terminal is shown below.

```
AttackBox Terminal

user@AttackBox# ssh sammie@10.49.146.171
The authenticity of host '10.49.146.171 (10.49.146.171)' can't be established.
ECDSA key fingerprint is SHA256:IFP+sTfHTDm72Ta2zfK9XjKASr30+ya4ic/ApEIziio.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.49.146.171' (ECDSA) to the list of known hosts.
sammie@10.49.146.171's password:
Welcome to Ubuntu 20.04.4 LTS (GNU/Linux 5.4.0-100-generic x86_64)
 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Tue 1 Mar 13:20:32 UTC 2022

System load:  0.03          Processes:            216
Usage of /:   51.8% of 6.53GB Users logged in:             1
Memory usage: 17%          IPv4 address for ens33: 10.49.146.171
Swap usage:   0%

* Super-optimized for small spaces - read how we shrank the memory
  footprint of MicroK8s to make it the smallest full K8s around.

https://ubuntu.com/blog/microk8s-memory-optimisation

0 updates can be applied immediately.

Last login: Tue Mar 1 09:46:11 2022 from 10.49.146.171
```

Amazing! It worked! Let's confirm that we are logged in as Sammie using the `whoami` (*who am I?*) command, which should return `sammie`.

To list the files in the current directory, we can use `ls`, short for *list*. This command will show all the files in the current directory unless they are hidden.

If you want to display the contents of any text file, you can use the command `cat FILENAME`, short for *concatenate*. This

command will print the contents of the file on the screen.

In the terminal below, we see the usage of the four commands: `ssh`, `whoami`, `ls`, and `cat`. Please follow along from the AttackBox's terminal.

```
AttackBox Terminal

user@AttackBox# ssh sammie@10.49.146.171
sammie@10.49.146.171's password:
Welcome to Ubuntu 20.04.4 LTS (GNU/Linux 5.4.0-100-generic x86_64)

[...]

Last login: Tue Mar  1 14:45:53 2022 from 10.20.30.1
sammie@beginner-os-security:~$ whoami
sammie
sammie@beginner-os-security:~$ ls
country.txt  draft.md  icon.png  password.txt  profile.jpg
sammie@beginner-os-security:~$ cat draft.md
# Operating System Security

Reusing passwords means that your password for other sites becomes exposed if one service is hacked.
sammie@beginner-os-security:~$
```

In our brief introduction to `Linux`, the last command that we will cover is `history`. This command prints the commands used by the user.

We have learned about two other usernames that can access the attached machine. They are:

- `johnny`
- `linda`

We know that both of these users have little regard for cybersecurity best practices. We can use several ways to guess the passwords for these two users. Here we list two approaches:

- If you are **not** logged in as `sammie` or any other user, you can use `ssh johnny@10.49.146.171` and manually try one password after the next to see which password works for `johnny`.
- If you are logged in as `sammie` or any other user, you can use `su - johnny` and manually try one password after the next to see which password works for `johnny`.

Answer the questions below

Based on the top 7 passwords, let's try to find Johnny's password. What is the password for the user `johnny`?

✓ Correct Answer



Once you are logged in as Johnny, use the command `history` to check the commands that Johnny has typed. We expect Johnny to have mistakenly typed the `root` password instead of a command. What is the root password?

✓ Correct Answer

While logged in as Johnny, use the command `su - root` to switch to the `root` account. Display the contents of the file `flag.txt` in the `root` directory. What is the content of the file?

THM{YouGotRoot}

✓ Correct Answer



Great work Abhishek26! Room completed!

Your skills are skyrocketing!

Completed tasks

✓ 3

Points earned

🎯 40

Streak

🔥 13